Data wrangling verbs

The native pipe `|>` *and the verbs in the* ***dplyr*** *package*

filter()
summarize()
group_by()
mutate()
arrange()
inner_join()
select()

The pipe operator

|>



Take this data frame, **then**…

|>



filter the data, **then** with those results…

|>



summarize

- “Pipe” a data frame into a “verb” command
- “Chain” the results from one “verb” command into another
- Think of it as the word **“then”**

filter()

Pull out just the **rows** you want in a data set, based on some criteria

Subset Observations (Rows)



filter()

Subset Observations (Rows)



Example

the data frame

```
on_time_flights <- flights |>
  filter(arr_delay < 30)
```

only keep rows where the column meets this criterion

the verb (function)

what column (variable) to use for filtering

filter()

Subset Observations (Rows)



Comparisons: `==` `!=` `<` `>` `<=` `>=`, combined with `&` (and) or `|` (or)

Another example

the data frame

```
newark_flights <- flights |>
  filter(origin == "EWR")
```

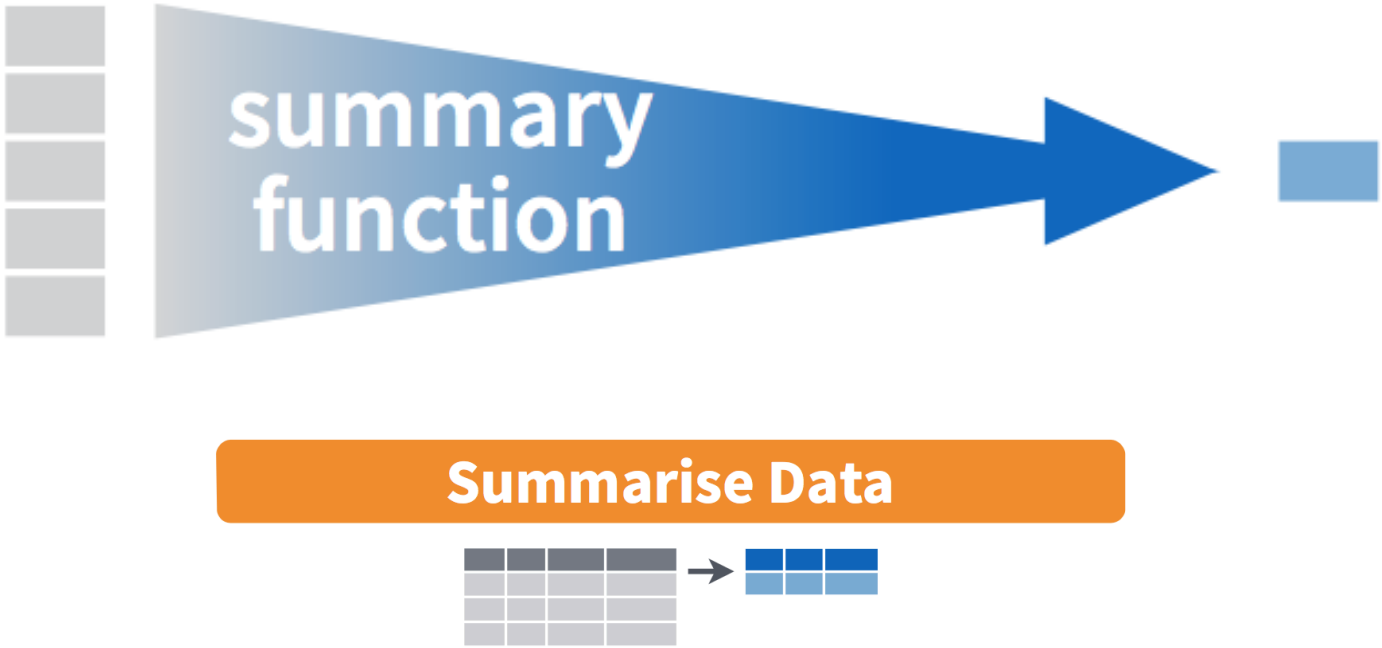
only keep rows where the column meets this criterion

the verb (function)

what column (variable) to use for filtering

summarize()

Take a column of data from a data frame and reduce it down to a single summary statistic



summarize()



Example

the data frame

```
flights |>
  summarize(av_dist = mean(distance))
```

what summary statistic to calculate

the verb (function)

what to call the result

the column to summarize

group_by() then summarize()

Take a column of data and reduce it down to a summary statistic, by some grouping variable



group_by() then summarize()



Example

the data frame

```
flights |>
  group_by(origin) |>
  summarize(av_dist = mean(distance))
```

group_by(), then what to group by

the verb (function)

what to call the result

the column to summarize

mutate()

Add a new column to a data frame, computed from the columns already there



mutate()



Example

the data frame

```
flights |>
  mutate(dist_km = distance * 1.609)
```

the verb (function)

what to call the new column (new variable)

what to put in the new column

arrange()

Sort the **rows** of a data frame by the values in one of its columns

Arrange Rows



arrange()

`freq_dest` has one row per destination and its number of flights, made with `group_by(dest)` and `summarize()`.

Example

the data frame

```
freq_dest |>
  arrange(desc(num_flights))
```

the verb (function)

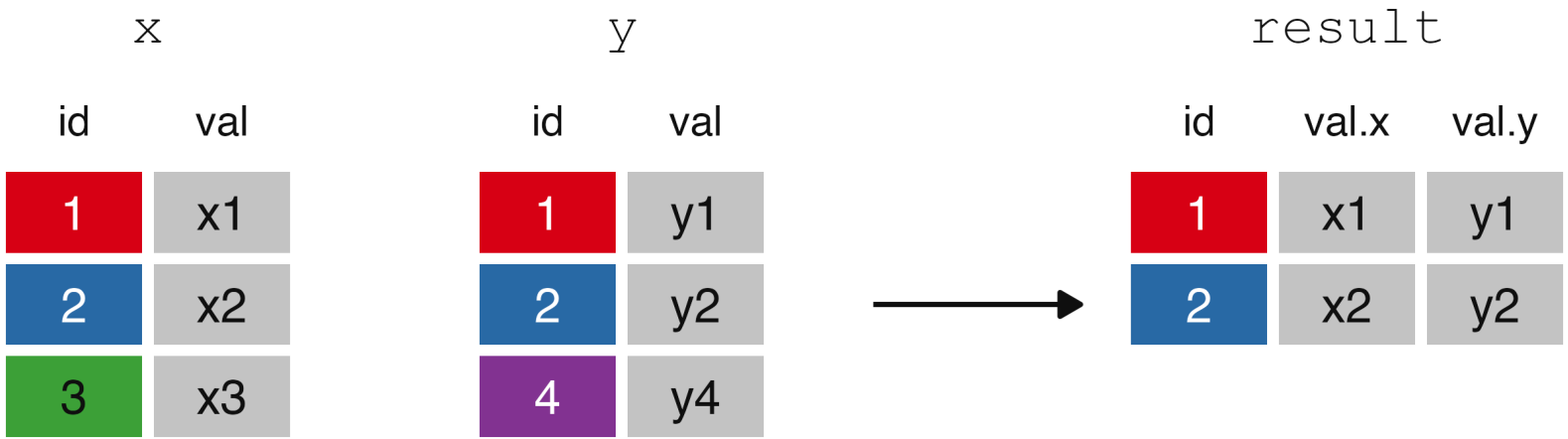
which column to sort by, largest first

inner_join()

Diagram adapted from Garrick Aden-Buie's tidyexplain

Combine two data frames by matching **rows** on a key column, keeping only the keys found in both

Join Data Frames



inner_join()

Example

the data frame

```
flights_joined <- flights |>
  inner_join(airlines, by = "carrier")
```

the verb (function)

the second data frame

the key column both share

select()

Pull out just the **columns** you want in a data set, based on the column names

Subset Variables (Columns)



select()

Subset Variables (Columns)



Example

the data frame

```
flights_sub <- flights |>
  select(arr_delay, dep_delay)
```

the verb (function)

what columns (variables) to keep